UHF Prime Reader
User Manual V1.0

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1.Docking Communication Protocol

The module can communicate with external application thruogh RS232.TTL,RS485.RJ45

1.1 Serial port communication parameter

Module use serial port communicate with external application

Sheet A-1 Module dock serial port parameter

Item	Parameter	Remarks
Baud Rate	115200bps	
Data bit	8	
Stop bit	1	
parity check bit	None	

1.2 Data Frame Format

The data packets sent by the upper-layer application are hereinafter referred to as "commands", and the data packets returned by the module to the upper-layer applications are hereinafter referred to as "responses". All the following data segments are in bytes in length.

Data transmission order: for each data item consisting of multiple bytes, the most significant byte is sent first, and the least significant bit within the byte is sent first.

The data frame format of command refer sheet A-2:

Sheet A-2: The data frame format of command

MSB					LSB
	Control Field			Information Field	EpLENogue Field
HEAD	ADDR	CMD	LEN	Data[]	CHECK
1Byte	1Byte	2Bytes	1Byte	0~255Byte	2Byte

The data frame format of response refer sheet A-3:

Sheet A-3: The data frame format of response

MSB						LSB
	Control Field			Information Field		Verify Field
HEAD	ADDR	CMD	LEN	STATUS	Data[]	CHECK
1Byte	1Byte	2Bytes	1Byte	1Byte	0~254Byte	2Byte

Data frame structure include three parts: Control Field、Information Field、Verify Field

		Length (Byte)	Specification
Control Field	HEAD	1	HEAD fixed as HEX data 0xCF， this byte use for data frmae synchronization
	ADDR	1	Reader address Address range:0x00~0xFE， 0xFF as broadcast address, reader only response to the command 0xFF that have same its own address, reader default address as 0x00
	CMD	2	Command code Command bit definition refere to sheet A-7
	LEN	1	Thie length is data length of all bytes of the actual Information Field, max data lengh is 255Bytes（0xFF）
Information Field	STATUS	1	STATUS is the execution status of the upper computer command contained in the module's response. STATUS only exists in the response data frame returned by the module, there is no STATUS part in the command frame sent by the upper-layer application. A STATUS of 0 indicates that the command was successfully executed. STATUS definition refer to sheet A-6.
	Data[]	NO	Parameter Field Under actual command,allow not exsit Data[] is actual data need transmit,will specific define Data[] valid byte under different kinds command format
Verify Field	MSB-CRC16	1	CRC16 high byte。

	LSB-CRC16	1	CRC16 low byte。	CRC16 value is all data from HEAD to Data[].CRC16 reference code refer appendix B

2 Command frame set

UHF RFID READER MODULE, below description call "module"

2.1 Command list

Sheet A-7 Example list of control command

Command Nmae	CMD Control byte	Function description	Remarks
EPC COMMAND			
RFM_INVENTORYISO_CONTINUE	0x0001	Tag inventory (Flows)	
RFM_INVENTORYISO_STOP	0x0002	Stop tag inventory	
MODULE DEFINTION COMMAND			
RFM_MODULE_INT	0X0050	Initialize device	
RFM_REBOOT	0x0052	Recover factory setting	Recover basic parameter default value
RFM_SET_PWR	0x0053	Set RF power	0-30dBm
RFM_SET_GET_RFIDPRO	0x0059	Set/read RF protocol	Only support ISO 18000-6C
RFM_SET_GET_NETPARA	0X005F	Set/read network information	Applicable to LAN
RFM_SET_GET_REMOTE_NETPARA	0x0064	Read obtain set remote network information	Applicable to the whole network, across network segments
RFM_GET_DEVICEINFO	0X0070	Obtain all module version NO and SN code	
RFM_SET_ALL_PARAM	0x0071	Set all base parameter	One time full setting
RFM_GET_ALL_PARAM	0x0072	Read all base parameter	One time full reading
RFM_SET_GET_PERMISSION_PARAM	0x0073	Read limits setting	
RFM_SET_GET_IOPUT_PARAM	0x0074	Input/output IO settting	
RFM_SET_GET_WiFi_PARAM	0x0075	Set/read WIFI information	
RFM_RELEASE_CLOSE_RELAY	0X0077	Release/close relay	

2.2 Module general control command

2.2.1 RFM_MODULE_INT（Initialize device ）

Confirm whether device online, it is used when upper computer connected device

- Command format

HEAD	ADDR	CMD	LEN	CHECK
0xCF	0xFF	0x0050	0x00	2Byte

- Response format and status byte

HEAD	ADDR	CMD	LEN	STATUS	CHECK
0xCF	0x00	0x0050	0x01	1Byte	2Byte

■ STATUS：This comand only return execution success (value 0*00), other value no valid

2.2.2 RFM_REBOOT（Recover factory setting）

Recover factory setting command, when that command executive success, module base parameter recover factory setting

- Command format

HEAD	ADDR	CMD	LEN	CHECK
0xCF	0xFF	0x0052	0x00	2Byte

- Response format and status byte

HEAD	ADDR	CMD	LEN	STATUS	CHECK
0xCF	0x00	0x0052	0x01	1Byte	2Byte

■ STATUS：0x00 executive success,other valve refere to sheet A-6

2.2.3 RFM_SET_PWR (Set RF output power)

This comamnd use to set module RF output power.before use module read tag need use that command to set module RF output power.If user don’ t set module RF output power, module will use default seeting when working

➢ Comamnd format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				Power	Resv	
0xCF	0xFF	0x0053	0x02	1Byte	1Byte	2Byte

■ Power：Output power,unit :dBm, value range：[0, 26]dBm，default: 26。

■ Resv：system reserved field,default:0x00；

➢ Response format and status byte

HEAD	ADDR	CMD	LEN	STATUS	CHECK
0xCF	0x00	0x0053	0x01	1Byte	2Byte

■ STATUS：0x00 execution success，other value refer to sheet A-6；

2.2.4 RFM_SET_GET_RFIDPRO (Set/read module support RF protocol standard)

This command set/read module RFID protocol standard (set protocol standards by running this command to check whether the module is successfully connected)

➢ Command format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				Option	RFIDPRO	
0xCF	0xFF	0x0059	N Byte	1Byte	NC or 1Byte	2Byte

■ Option：command control options

0x01：Set, follow 1 byte length RFIDPRO 设置；

0x02：Read, no follow RFIDPRO

Other value: No valid；

■ RFIDPRO：protocol options, details refer to below sheet:

➢ Response format and status byte

Set response：

HEAD	ADDR	CMD	LEN	STATUS	Option	CHECK
0xCF	0x00	0x0059	1Byte	1Byte	1Byte	2Byte

Obtain response:

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD		CHECK
					Option	RFIDPRO	
0xCF	0x00	0x0059	3Byte	1Byte	1Byte	1Byte	2Byte

Field	Byte	Description
STATUS	1	0x00 execution success other value refer to sheet A-6
Option	1	0x01: set; 0x02: obtain
RFIDPRO	1	0x00: ISO 18000-6C; 0x01: GB/T 29768; 0x02: GJB 7377.1; only support ISO 18000-6C。

2.2.5 RFM_SET_GET_NETPARA (set/read device network information)

This command set/read module network information, set default as TCP Server, upper computer connect set related device IP address and port

- Comamnd format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				Option	NetPara	
0xCF	0xFF	0x005F	1 or 2 Byte	1Byte	NC or 1Byte	2Byte

  - Option: Command control options

    0x01: set;

    0x02: obtain;

    Ohter value: no valid;

- Response format and status byte

Set response:

HEAD	ADDR	CMD	LEN	STATUS	Option	CHECK
0xCF	0x00	0x005F	1Byte	1Byte	1Byte	2Byte

Obtain response:

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD		CHECK
					Option	NetPara	
0xCF	0x00	0x005F	2+N Byte	1Byte	1Byte	N Byte	2Byte

Field		Byte	Description
STATUS		1	0x00 execution success other value refere to sheet A-6;
Option		1	0x01: set ; 0x02: obtain
NetPara	IPAddr	4	Set IP address, if IP address as 192.168.1.1, data as [0xC0,A8,0x01,0x01]
	MacAddr	6	Mac addres, if multiple device work at same potion, please do not use same MAC address : Example: mac[6]={0x00,0x08,0xdc,0x11,0x11,0x11} Mac address as 00-08-DC-11-11-11
	Port	2	Listening port, value as [0,65536], default 2022
	NetMask	4	Net mask code , default as [0xFF,0xFF,0xFF,0x00]
	GateWay	4	Default gataway,default [0xC0,0xA8,0x01,0x01]

2.2.6 RFM_SET_GET_REMOTE_NETPARA (set/read remote network information)

This command set/read remote network information, device sent data to remote IP, remote PC can listening and receiving data

- Comamnd format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				Option	NetPara	
0xCF	0xFF	0x0064	1 or 2 Byte	1Byte	NC or 1Byte	2Byte

- Option：Command control options

  0x01：set；

  0x02：obtain；

  Ohter value：no valid；

➢ Response format and status byte

Set response：

HEAD	ADDR	CMD	LEN	STATUS	Option	CHECK
0xCF	0x00	0x0064	2Byte	1Byte	1Byte	2Byte

Obtain response：

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD		CHECK
					Option	NetPara	
0xCF	0x00	0x0064	2 + N Byte	1Byte	1Byte	NC or N Byte	2Byte

Field		Byte	Description
STATUS		1	0x00 execution success other value refer to seet A-6；
Option		1	0x01：set； 0x02：Obtain
NetPara	Enable	1	Enable switch, 1: enable sent,0: close send
	IPAddr	4	Remote IP address，if IP address as 192.168.1.1,data will be [0xC0,0xA8,0x01,0x01]
	Port	2	Listening port,value as [0,65536]，default 5000；

	HeartTime	1	Heartbeat packet duration to prevent dropped calls (unit: 5S)

2.2.7 RFM_GET_DEVICEINFO (Obtain module information)

This command use to obtain current device version information,include CP module and RFID module hardware version number, firmware version and SN number

- Command format

HEAD	ADDR	CMD	LEN	CHECK
0xCF	0xFF	0x0070	0x00	2Byte

- Response format and status byte

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD		
					HardVer	FirmVer	SN_code
0xCF	0x00	0x0070	0x99	1Byte	32Bytes	32Bytes	12Bytes

All bytes definition:

PAYLOAD			CHECK
Reserve-I	Reserve-II	Reserve-III	
32Bytes	32Bytes	12Bytes	2Byte

Field	Byte	Description
STATUS	1	This command only return execution success (value 0*00), other value no valid
HardVer	32	Module hardware version NO, format as ASCII code, hardware version NO short of 32 byte, balance other byte use” 0” supply
FirmVer	32	Module firmware version NO, format same as hardware version NO
SNCode	12	Module 12 bytes serial number, format as ASCII code
Reserve-I	32	Reserve

Reserve-II	32	Reserve
Reserve-III	12	Reserve

2.2.8 RFM_SET_ALL_PARAM (Set all configurable parameter)

This command use to set device all configurable parameter, before use module read tag need use that command to set module RF output power.If user don’ t set module RF output power, module will use default setting when working

➢ Command format

HEAD	ADDR	CMD	LEN	PAYLOAD							
				Addr	RFIDPRO	Work Mode	Interface	Baudrate	WGSet	Ant	RfidFreq
0xCF	0xFF	0x0071	N Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	8Bytes

PAYLOAD										CHECK
RfidPower	InquiryArea	QValue	Session	AcsAddr	AcsDataLen	FilterTime	TriggerTime	BuzzerTime	Polling Interval	
1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	2Bytes

- Addr：device communication address, default as 0x00, this address can not be 0xFF, if set as 0xFF, module will return parameter error information
- RFIDPRO：Device RFID protocol standard specification:0x00：ISO 18000-6C；0x01：GB/T 29768；0x02：GJB 7377.1；at present only support ISO 18000-6C。
- WorkMode：device work mode, default 0

WorkMode	Work mode
0	Answer mode
1	Active mode
2	Trigger mode

- Interface：device communication interface, default 0x80,specific definition:

Interface	Type
0x80	RS232
0x40	RS485
0x20	RJ45
0x10	WiFi

- Baudrate：baud rate, default value as 4, specific defintion:

baudrate	Actual baud rate
0	9600bps
1	19200 bps
2	38400 bps
3	57600 bps
4	115200 bps

- WGSet：wiegand data output interface configuration parameter, default value 0x00,specific defintion as below:

WGSet	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Bit definition	0：close WG output 1：open WG output	0：wg26 1：wg34	0：low bit first 1：high bit first	Reserved	Reserved	Reserved	Reserved	Reserved

- Ant：device all antenna NO, use bit as selected antenna, related bit value as 1 means using that antenna, value as 0 means don’ t use that antenna. Start from low bit, NO 0 means NO 1 antenna, first bit means NO 2 antenna and so on, max can be 8 antenna, different module support different qty antenna, depends on actual sitiation, default value as 0X01, means NO 1 antenna

- RfidFreq：device RFID freqeuncy related parameter, use to select every frequency range upper limit freqeuncy points and lower limit freqeuncy points, lenth is 8 bytes, specific parameter as below:

RfidFreq				
REGION	STRATFREI	STRATFRED	STEPFRE	CN
1 Byte	2Byte	2Byte	2Byte	1Byte

Every bytes definition：

Set freqeuncy according REGION confirmed freqeuncy range, then refer STRATFREI、STRATFRED、STEPFRE、CN calculate specific frequency range

Field	byte	Description
REGION	1	Each country frequency range ： 0x00: User define according demand； 0x01：US [902.75~927.25] 0x02：Korea [917.1~923.5] 0x03：EU [865.1~868.1] 0x04：JAPAN [952.2~953.6] 0x05：MALAYSIA [919.5~922.5] 0x06：EU3 [865.7~867,5] 0x07：CHINA_BAND1 [840.125~844.875] 0x08：CHINA_BAND2 [920.125~924.875]
STRATFREI	2	MHz start freqeuncy integer part: for example 920.125MHz,STRATFREI = 920 = 0x0398,high byte=0x03, low byte=0x98；
STRATFRED	2	MHz start freqeuncy fractional part,for example 920.125MHz，STRATFRED =125, high byte=0x00,low byte=0x7D

STEPFRE	2	Frequency step (KHz), refer to the calculation formula of each frequency band; For example, 125KHz, STEPFRE =125, high byte =0x00, low byte =0x7D;
CN	1	Channel number

Frequency computational formula:

Min Frequency point: Fmin = STRATFREI + STRATFRED/1000 (Unit: MHz)

Max Frequency point: Fmax = Fmin + STEPFRE*CN/1000 (Unit: MHz)

Calculation formula for each frequency band :(the frequency range shall be within the national standard)

Chinese band2: Fs = 920.125 + CN * 0.25 (MHz) CN∈[0, 19]。
US band: Fs = 902.75 + CN * 0.5 (MHz) CN∈[0,49]。
Korean band: Fs = 917.1 + CN * 0.2 (MHz) CN∈[0, 31]。
EU band: Fs = 865.1 + CN*0.2(MHz) CN∈[0, 14]。
Ukraine band: Fs = 868.0 + CN*0.1(MHz) CN∈[0, 6]。
Peru band : Fs = 916.2 + CN*0.9(MHz) CN∈[0, 11]。

Chinese band1: Fs = 840.125 + CN * 0.25 (MHz) CN∈[0, 19]。
EU3 band: Fs = 865.7 + CN * 0.6(MHz) CN∈[0, 3]。
US band3: Fs = 902 + CN * 0.5 (MHz) CN∈[0,52]。
Taiwan band: Fs = 922.25 + CN * 0.5 (MHz) CN∈[0,11]。

- RfidPower: RFID output power of equipment, unit: DBM, value range: [0, 30] DBM, other invalid.
- InquiryArea: The storage area of the tag to be accessed by the device. 0x00: reserved area; 0x01 (default): EPC storage area; 0x02: TID storage area; 0x03: user storage area; Other values are reserved. If there are other values in the command, the message of parameter error will be returned.
- QValue: The initial Q value used when querying EPC tags shall be set so that the number of tags in the field is about 2q. The default value of Q value is 4, and the

range of Q value is 0-15. If other values appear in the command, a parameter error message will be returned.

- Session：The session value used when querying the EPC tag. The default value is 0. The value range is [0, 3]. Other values will return the message of parameter error.

Session	Notes
0	Session use S0
1	Session use S1
2	Session use S2
3	Session use S3

- AcsAddr：Device access the starting address of the label storage area

Unit: Byte, Default value:0x00: When accessing the EPC area, 0x00 indicates the starting address of the EPC number segment of the EPC area except CRC and PC segment; When accessing other storage areas, 0x00 indicates the starting address of the storage area.

- AcsDataLen：Device access the data length of the label storage area, Unit: Byte, Default value:0x00。
- FilterTime：Filter time, after reading the label successfully within the value time ; Filter out tags with the same data. Unit: s, value range: [0, 255], others are invalid; The default value is 0, and there is no filtering.
- TriggerTime：The enquiry time after the equipment receives the trigger signal. The unit is S, the default value is 1, the value range is [0, 255], and the others are invalid.
- BuzzerTime：Buzzer duration after equipment execution successfully，Unit: 10ms, value range: [0, 255], others are invalid; The default value is 1. When it is 0, the buzzer does not sound。
- PollingInterval：Query interval, unit: 10ms, value range: [0, 255], others are invalid, and the default is 1.

➢ Response format and status bytes

HEAD	ADDR	CMD	LEN	STATUS	CHECK
0xCF	0xXX	0x0071	0x01	1Byte	2Byte

- STATUS:

1) 0x00: Command execution succeeded;
2) 0x01: Parameter error;
3) Other values: invalid;

2.2.9 RFM_GET_ALL_PARAM

This command is used to read the RF power currently set by the module. The user can use this command to view the power settings of the module before using the module to operate with the tag.

- Command format

HEAD	ADDR	CMD	LEN	CHECK
0xCF	0xFF	0x0072	0x00	2Byte

- Response format and status byte

HEAD	ADDR	CMD	LEN	PAYLOAD								
				STATUS	Addr	RFIDPRO	Work Mode	Interface	Baudrate	WGSet	Ant	RfidFreq
0xCF	0xFF	0x0071	0x1A	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	8Bytes

PAYLOAD										CHECK
RfidPower	InquiryArea	QValue	Session	AcsAddr	AcsDataLen	FilterTime	TriggerTime	BuzzerTime	Polling Interval	
1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	2Bytes

- STATUS:

1) 0x00: Command execution succeeded;
2) Other values: invalid;

- Other parameters please refer to RFM_SET_ALL_PARAM。

2.2.10 RFM_SET_GET_PERMISSION_PARAM

This command is used to set password and mask query permission parameters, relay usage parameters, cache switch parameters and public protocol usage parameters.

- Command format

HEAD	Control Field			PAYLOAD								CHECK
	ADDR	CMD	LEN	OPTION	Code EN	Codes	Mask EN	Start Add	Mask Len	Mask Data	Mask Condition	
0xCF	0xFF	0x0073	0x16	1Byte	1Byte	4Bytes	1Byte	1Byte	1Byte	12Bytes	1Byte	2Bytes

Parameter definition:

Option: Command control options, 0x01 set, 0x02 read, other values are invalid.

CodeEN: Password function parameter, length is 1 byte. 0x01 enabled, 0x00 not enabled. Default 0x00。

Codes: The access password of the tag is 4 bytes, default[0x00,0x00,0x00,0x00]。

MaskEN: Mask function parameter, length is 1 byte. 0x01 enabled, 0x00 not enabled. The default is 0x00.

StartAdd: Mask start address, length is 1 byte, unit is byte. The default is 0x00.

Mask Len: Mask length, length is 1 byte, unit is byte, maximum 12. The default is 0x00.

MaskData: The mask data is 12 bytes, when the mask length is less than 12 bytes, the following byte data is supplemented with 0.

Default [0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00]。

MaskCondition: Mask condition, length is 1 byte. 0x00: the password or mask matches; 0x01: the password and mask match at the same time. The default is 0x00.

- Response format and status bytes

Set response:

HEAD	ADDR	CMD	LEN	STATUS	Option	CHECK
0xCF	0x00	0x0073	1Byte	1Byte	1Byte	2Byte

Get response:

HEAD	Control Field	PAYLOAD	CHECK

	ADDR	CMD	LEN	STATUS	OPTION	Code EN	Codes	Mask EN	Start Add	Mask Len	Mask Data	Mask Condition	
0xCF	0xFF	0x0073	0x17	1Byte	1Byte	1Byte	4Byte	1Byte	1Byte	1Byte	12Bytes	1Byte	2Bytes

- STATUS:

3) 0x00: Command execution succeeded;

4) Other values: invalid;

2.2.11 RFM_SET_GET_IOPUT_PARAM

This command is used to read the advanced parameters of the device. Please see the specific analysis of the parameters 2.2.9。

- Command format

HEAD	Control Field			PAYLOAD									CHECK
	ADDR	CMD	LEN	OPTION	Relay EN	Relay Time	Relay Power EN	Trig Way	Buffer EN	Protocol EN	Protocol Type	Protocol Format	
0xCF	0xFF	0x0074	0x12	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	10 Bytes	2Bytes

Parameter definition:

Option: Command control options, 0x01 set, 0x02 read, other values are invalid.

RelayEN: Relay function parameter, length is 1 byte. 0x01 enabled, 0x00 not enabled. The default is 0x00.

RelayTime: Relay normally open time parameter, length is 1 byte, unit is S. The default is 0x03.

RelayPowerEN: Relay power on parameter, length is 1 byte. 0x01 is on, 0x00 is not on. The default is 0x00.

TrigWay: The effective trigger mode parameter, length is 1 byte. 0x01 high level (2v-5v) trigger, 0x00 low level (less than 0.5V) trigger. The default is 0x01.

BufferEN: Cache function parameter, length is 1 byte. 0x01 enabled, 0x00 not enabled. The default is 0x00.

ProtocolEN: Specific data interaction function parameter, length 1 byte. 0x01 enabled, 0x00 not enabled. The default is 0x00.

ProtocolType: Specific data interaction protocol type, length is 1 byte. The default is 0x00.

ProtocolFormat: Specific data interaction protocol format, length is 10 bytes.

default[0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00]。

➢ Response format and status byte

Set response:

HEAD	ADDR	CMD	LEN	STATUS	Option	CHECK
0xCF	0x00	0x0073	1Byte	1Byte	1Byte	2Byte

Get response:

HEAD	Control Field				PAYLOAD										CHECK
	ADDR	CMD	LEN		STATUS	OPTION	Relay EN	Relay Time	Relay Power EN	Trig Way	Buffer EN	Protocol EN	Protocol Type	Protocol Format	
0xCF	0xFF	0x0074	0x13		1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	1Byte	10 Bytes	2Bytes

■ STATUS:

1) 0x00: Command execution succeeded;

2) Other values: invalid;

2.2.12 RFM_SET_GET_WiFi_PARAM

This command sets / reads the WiFi information of the module, sets it to TCP client, and needs to set the corresponding remote IP address and port; WiFi distribution mode: 1smartconfig distributes the network through the mobile app, 2. Serial port distribution, which is configured by sending SSID and PWD. Power on and connect automatically after the configuration is successful.

➢ Command format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				Option	WiFiPara	
0xCF	0xFF	0x0075	1Byte	1Byte	NC or 1Byte	2Byte

■ Option: Command control options

0x01：Set；

0x02：Read；

Other values: invalid;

- Response format and status byte

Set response:

HEAD	ADDR	CMD	LEN	STATUS	Option	CHECK
0xCF	0x00	0x0075	1Byte	1Byte	1Byte	2Byte

Get response:

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD		CHECK
					Option	NetPara	
0xCF	0x00	0x0075	1Byte	1Byte	1Byte	NByte	2Byte

Filed		Byte	Description
STATUS		1	0x00 is executed successfully, other values are shown in Table A-6;
Option		1	0x01: setting; 0x02: get
WiFiPara	WiFiEN	1	WiFi parameters, mode selection. Bit0:0: turn off WiFi; 1: Turn on WiFi Bit1: 0: smartconfig 1: serial port distribution network. Other bits default to 0;
	SSID	32	WiFi name, ASIC code. Invalid bytes are filled with 0x00

	PWD	64	WiFi password, ASIC code. Invalid bytes are filled with 0x00
	RemoteIPAddr	4	Remote IP address. If the IP address is 192.168.1.1, the data is [0xC0,A8,0x01,0x64]
	RemotePort	2	Remote IP port number, high byte in front, low byte after, Default: 0x1388 (5000)

2.2.13 RFM_RELEASE_CLOSE_RELAY

- This command is used to release and close the relay.
- Command format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				Option	Valid'Time	
0xCF	0xFF	0x0077	1Byte	1Byte	1Byte	2Byte

  - Option：command control options

    0x01：release；

    0x02：close；

    Other values: invalid；

  - ValidTime：The effective time when closing, the unit is S, the default value is 1. When the ValidTime value is equal to 0, the relay will always be in the closed state; when the ValidTime value is greater than 0, the relay will be released after closing ValidTime seconds.

- Response format and status bytes

HEAD	ADDR	CMD	LEN	STATUS	Option	Valid'Time	CHECK
0xCF	0x00	0x0077	1Byte	1Byte	1Byte	1Byte	2Byte

Field	byte	Describe
STATUS	1	0x00: The command is executed successfully; 0x01: The parameter is wrong.
Option	1	0x01: release; 0x02: close
ValidTime	1	The effective time when closing, the unit is S, the default value is 1.

2.3 International standard (ISO 18000-6c) protocol related command

2.3.1 RFM_INVENTORYISO_CONTINUE

This command is to start the international standard multi label counting command, and the counting function has an anti-collision algorithm.

This command can specify the counting duration. If the number of counts entered is 0, it means that the counting label will continue until the stop counting command is received. During the counting process, every time a label is counted successfully, the label information of the new inventory point will be returned through a command response with a "status" value of "0".

When the counting command is successfully executed, the command response with the value of "status" of "0x12" will be returned at last to inform that the counting command has been executed.

- Command format

HEAD	ADDR	CMD	LEN	PAYLOAD		CHECK
				InvType	InvParam	
0xCF	0xFF	0x0001	0x05	1Byte	4Byte	2Byte

- InvType：Counting method:

  0x00：Counting tag according to the time, stop counting after executing the specified time or stop counting after receiving the stop counting command;

  0x01: Count according to the number of cycles, and stop counting after executing the specified number of polls or receiving the stop counting instruction;

- InvParam ： Counting method parameters:

  1.If InvType is 0x00：

  InvParam indicates the counting time, unit is seconds. If the value is 0, it indicates that the counting label will continue until the stop counting command is received;

  2.If InvType is 0x01：

  InvParam indicates the number of counts, unit is times. The value must be greater than 0;

➢ Response format and status bytes

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD					CHECK
					RSSI	Antenna	Channel	EPC LEN	EPC NUM	
0xCF	0x00	0x0001	1Byte	1Byte	2Bytes	1 Byte	1 Byte	1Byte	N Bytes	2Byte

各字节定义：

Filed	Byte	Description

STATUS	1	0x00：The tag is counted successfully, and PAYLOAD contains the tag information; 0x01：Q、MemBank parameter value is wrong or the length is inconsistent with the mask data length; 0x02：Command execution failure due to module internal error; 0x12：There is no counting of the label or the entire counting order is completed; 0x17： The tag data exceeds the maximum transmission length of the serial port; Other values: invalid;
RSSI	2	The RSSI of the tag ACK response, unit is 0.1dBm, signed numbers, negative numbers use complement format;
Antenna	1	The antenna that reads the data, bit0 represents antenna 1, and so on.
Channel	1	The data of label received from which channel, the value starts from 0, 0 means 0 channel, 1 means 1 channel, and so on;
EPC LEN	1	EPC number length of label (bytes);
EPC NUM	N Byte	EPC number of the label;

2.3.2 RFM_INVENTORY_STOP

This command the user to actively stop the international standard multi label anti-collision inventory process.

- Command format

HEAD	ADDR	CMD	LEN	CHECK
0xCF	0xFF	0x0002	0x00	2Byte

➢ Response format and status byte

HEAD	ADDR	CMD	LEN	STATUS	CHECK
0xCF	0x00	0x0002	0x01	1Byte	2Byte

■ STATUS：This command will only return successful execution (the value is 0x00), and other values are invalid;

2.3.3 RFM_INVENTORY_ACTIVE (Active mode)

In active mode, the uploaded data format.

HEAD	ADDR	CMD	LEN	STATUS	PAYLOAD					CHECK
					RSSI	Antenna	Channel	Data LEN	Data	
0xCF	0x00	0x0001	N Byte	1Byte	2Bytes	1 Byte	1 Byte	1Byte	N Bytes	2Byte

Definition of each byte:

Field	Byte	Describe
STATUS	1	0x00 is executed successfully, other values are shown in Table A-6;
RSSI	2	The RSSI of the tag ACK response, the unit is 0.1dBm, the number is signed, and the negative number uses the complement format;
Antenna	1	The antenna that reads the data, bit0 represents antenna 1, and so on.

Channel	1	From which channel the tag data is received, the value starts from 0, 0 means 0 channel, 1 means 1 channel, and so on;
Data LEN	1	Label's number length (bytes);
Data	N Byte	Tag data;

For example: read data (hex) as e2 80 6f 12 00 00 00 02 21 50 60 94

Rssi(hex)：fe 48 The actual complement is -44dbm

Data Frame(hex)：0xCF 00 00 01 12 00 fe 48 01 00 0c e2 80 6f 12 00 00 00 02 21 50 60 94 7b 78

Appendix A Operation status returned by label

The operation status code returned by the tag is 8 bits in total, see table A-1.

Table A-1 operation status of label return

Operation status code	Operation status	Description	Error priority
0x81	Other errors	Unknown error returned by label	
0x82	Storage overflow	The destination store does not exist	
0x83	Store lock	Write to the storage area locked as non writable, Perform or erase operation, and read the storage area locked as unreadable	
0x84	Insufficient power	The tag does not have enough energy to complete the operation	
0x85	Unspecified error	Unknown error returned by label	

Appendix B CRC16 check reference C code

Example:

```
#define PRESET_VALUE 0xFFFF
#define POLYNOMIAL   0x8408
unsigned int uiCrc16Cal(unsigned char const  * pucY, unsigned char ucX)
```

```
{
    unsigned char ucI,ucJ;
    unsigned short int   uiCrcValue = PRESET_VALUE;

        for(ucI = 0; ucI < ucX; ucI++)
        {
            uiCrcValue = uiCrcValue ^ *(pucY + ucI);
            for(ucJ = 0; ucJ < 8; ucJ++)
           {
             if(uiCrcValue & 0x0001)
             {
                 uiCrcValue = (uiCrcValue >> 1) ^ POLYNOMIAL;
             }
             else
             {
                 uiCrcValue = (uiCrcValue >> 1);
             }
        }
    }
     return uiCrcValue;
}
```

Appendix C Definition of status.

Table A-6 status definitions

STATUS	Error description
0x00	Successful execution (this only means that the module successfully received the tag response data. If there is a tag execution status in the tag response, you should further judge whether the tag execution status is correct)
0x01	The parameter value is wrong or out of range, or the module does not support the parameter value
0x02	Command execution failed due to module internal error

0x03	Reserve
0x12	There is no counting to the label or the entire counting command is completed
0x14	Label response timeout
0x15	Demodulation tag response error
0x16	Protocol authentication failed
0x17	Password error
0xFF	No more data